PROJECT DOCUMENTATION

Secure Family Private Cloud
& Learning Lab

Build a homelab that protects family data, provides private services, and develops practical IT skills through real hands-on experience.

PROJECT	FOCUS AREAS	USERS
Home Infrastructure	Storage · Backups · Self-Hosting · Security · IT Learning	Self & Family

A secure private cloud with backups, self-hosted services, monitoring, secure remote access, and a dedicated IT learning lab.

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1. Project Overview

This project documents the design, planning, and phased implementation of a personal homelab infrastructure. The system is built to serve two complementary purposes: providing a reliable, secure private cloud environment for the family, and acting as a hands-on IT learning platform for practical skill development in line with BTS studies and future career goals.

	Secure Family Private Cloud + Learning Lab
Main Focus	Storage, backup, self-hosting, monitoring, network security, and IT learning
	Self and family members
Long-Term Value	Practical home infrastructure and an active learning platform supporting BTS studies and future IT career development

KEY PRINCIPLE

The homelab follows a strict separation rule: stable family services are always isolated from the experimental lab side. Data safety comes first — no test environment can interfere with production files or core services.

2. Main Objectives

The homelab is driven by seven core objectives. Each one addresses a specific gap in the current home IT setup — from fragmented file storage and zero backup strategy to the need for hands-on technical experience beyond classroom theory.

1	**Centralized Storage** Store family photos, videos, certificates, diplomas, school files, and shared documents in one organized, private location. Replace scattered external drives and cloud subscriptions with a single, self-managed storage system.
2	**Reliable Backups** Protect all important data against disk failure, accidental deletion, file corruption, and catastrophic loss. Implement a structured backup strategy (e.g., 3-2-1 rule) to ensure recoverability in any scenario.
3	**Self-Hosting** Run private applications, dashboards, personal websites, and internal services from home. Gain full control over the software stack without relying on third-party cloud platforms.
4	**Secure Remote Access** Reach files and self-hosted services safely from any location — whether from school, work, or travelling — using encrypted, authenticated connections.
5	**System Monitoring** Track server health, storage utilization, service uptime, disk temperature, and overall system status. Receive alerts before problems escalate and maintain visibility at all times.
6	**Network Security & Privacy** Strengthen the home network with improved access controls, DNS filtering, firewall rules, and better protection for all family devices and data flows.
7	**IT Learning Platform** Use the homelab as a live, practical learning environment for virtualization, Linux administration, Docker, networking, security, backup strategy, and real-world troubleshooting.

3. Planned Homelab Structure

The homelab is built around a single main server running multiple logically separated services. This approach keeps the system manageable while maintaining clear boundaries between stable family infrastructure and experimental lab environments.

1. NAS / Storage Service

Central file storage and backup destination

- Family photos, videos, and media library
- Important documents, certificates, diplomas
- Shared family folders with access control
- Primary destination for all backup jobs
- Organized archive with folder structure

2. App Hosting Service

Self-hosted apps, websites, and dashboards

- Personal and family web applications
- Private dashboards and internal tools
- Personal websites and web projects
- Reverse proxy with SSL for all services
- Containerized deployment via Docker

3. Monitoring Service

System health, uptime, and resource tracking

- Real-time server and service uptime checks
- CPU, RAM, and disk usage dashboards
- Disk health monitoring (SMART data)
- Alert notifications for failures or anomalies
- Centralized logging and event history

4. Learning Lab

Test environments for IT practice and study

- Linux virtual machines for administration practice
- Windows environments for compatibility testing
- Networking labs and protocol experiments
- BTS project environments and coursework

- Fully resettable — safe to break and rebuild

5. Security / Firewall Layer

Network protection, access control, privacy

- Network-level firewall and traffic filtering
- DNS filtering and ad/tracker blocking
- VPN for encrypted remote access
- Segmented VLANs for device isolation
- Access control lists and intrusion detection

4. Core Design Principles

The homelab is governed by one key architectural rule: stable, family-critical services must always be isolated from experimental and test environments. This ensures data safety, system reliability, and uninterrupted access to important resources — even when the lab side is being tested, reconfigured, or deliberately broken.

Production services — always available

- Centralized storage for all family files
- Automated and verified backup jobs
- Shared folders and user access control
- Core self-hosted services (media, docs)
- System monitoring and alerting
- Firewall, DNS filtering, and VPN

LAB SIDE

Test environments — safe to experiment

- Linux virtual machines for admin practice
- Windows Server test environments
- Network protocol labs and experiments
- Docker and app deployment testing
- BTS project work and course labs
- Systems that can be freely broken and rebuilt

WHY THIS MATTERS

Any configuration mistake, failed experiment, or broken VM on the lab side must never be able to affect the family storage, backups, or running services. Isolation is enforced at the network, storage, and virtualization levels.

5. Build Phases

The homelab is built incrementally across five phases. Each phase delivers tangible value before the next begins, reducing risk and ensuring the core infrastructure is solid and tested before further complexity is introduced.

#	Phase	Main Work & Deliverables
1	**Useful Core**	Set up the main server, configure storage volumes and shared folders, create user accounts, establish network shares, and deploy the first backup strategy. The homelab becomes immediately useful to the family from day one.
2	**Self-Hosting**	Deploy app hosting infrastructure with Docker Compose, configure a reverse proxy with SSL (Nginx Proxy Manager or Caddy), and launch the first self-hosted services such as Nextcloud, Vaultwarden, or a personal website.
3	**Monitoring**	Deploy a full monitoring stack (Grafana + Prometheus, or Uptime Kuma) for service uptime, hardware health, disk usage, and resource consumption. Configure alerting for failures and anomalies to maintain full visibility.
4	**Security & Remote Access**	Harden the firewall ruleset, configure a VPN server for secure remote access, add DNS filtering for the home network, segment devices using VLANs, and review and tighten all access controls across all services.
5	**Learning Expansion**	Provision isolated virtual machines for Linux and Windows practice, create networking lab environments, set up dedicated BTS project workspaces, and build out the full structured learning platform with reusable templates.

6. Expected Benefits

The homelab delivers value across three dimensions: immediate benefits for the family at home, accelerated skill development for IT studies and career preparation, and a long-term infrastructure asset that grows with future needs and ambitions.

For Home & Family	For Learning & Growth	For the Future
✓ Safer, private storage for all family data	✓ Real hands-on practice beyond theory	✓ Strong portfolio and interview project
✓ Single location for all important documents	✓ Virtualization and Linux administration	✓ Foundation for advanced infrastructure
✓ Organized shared folder structure	✓ Docker, networking, and security work	✓ Platform for future apps and websites
✓ Backups that prevent permanent data loss	✓ Practical BTS project platform	✓ Skill base for professional certifications
✓ Private services with no third-party access	✓ Troubleshooting in live environments	✓ Infrastructure that scales with needs

7. Final Vision

The finished homelab brings together all five service layers into a coherent, dependable home infrastructure — something that actually works daily for the family while providing a live, evolving learning environment. This is not a one-time build but an ongoing project that grows in sophistication as skills and requirements develop.

A secure family private cloud with reliable backups, self-hosted services,

live monitoring, secure remote access, and a dedicated learning lab for real-world IT practice.

TECHNOLOGIES & TOOLS IN SCOPE	
Virtualization Proxmox VE or similar hypervisor	**Storage / NAS** TrueNAS, OpenMediaVault, or ZFS
Self-Hosting Docker, Docker Compose, Nginx Proxy Manager	**Monitoring** Grafana, Prometheus, Uptime Kuma
Security / Firewall pfSense, OPNsense, or OpenWRT	**VPN / Remote Access** WireGuard or OpenVPN
DNS Filtering Pi-hole or AdGuard Home	**OS / Platforms** Debian, Ubuntu Server, Windows Server